Assignment 1 (Due August 8, 2026)

Answer the following questions and submit the copies (soft copies) to this drive link:
https://drive.google.com/drive/folders/1t0ghr29wgLPoCVidzFHrQOqA30P-zrNt?usp=sharing

Theoretical questions

1. Explain the characteristics and importance of algorithms.
2. Distinguish between algorithms, flowcharts, pseudocode, and programs.
3. Discuss various symbols with labeled diagrams.

Sequence type.

1. Write an algorithm and draw the flowchart to find the area and perimeter of the circle.
2. Write an algorithm and draw a flowchart to convert temperature from Celsius to Fahrenheit. and vice versa.
3. Write an algorithm and draw the flowchart to determine the Simple Interest, reading Principle, Rate and Time.
4. Write algorithms and flowcharts:
    a. Using a third variable
    b. Without using a third variable

Conditional type.

5. Write an algorithm and draw the flowchart to check whether number is even or odd.
6. Write an algorithm and draw the flowchart to check whether given year is a leap year or not.
7. Write an algorithm and draw the flowchart to check if a triangle is equilateral triangle.
8. Write an algorithm and draw the flowchart to find largest of three numbers.
9. Write an algorithm and draw the flowchart to read the marks (0-100) and display the grade. (Grades: A(90-100), B(75-89), C(60-74), F(Below 60).

Repetition type

10. Write an algorithm and draw the flowchart to find factorial of a number
11. Write an algorithm and draw the flowchart to check if a number is palindrome or not.
12. Write an algorithm and draw the flowchart to find sum of digits of a given number.
13. Write an algorithm and draw the flowchart to check if the entered phone number is 10 digit.
14. Write an algorithm and draw the flowchart to find sum of first N natural numbers.
15. Write an algorithm and draw the flowchart to Generate the multiplication table for a given number up to 10.
16. Write an algorithm and draw the flowchart to Generate the first N terms of the Fibonacci series.

Using all.

17. Design an algorithm and flowchart for an ATM system. Include appropriate decision blocks.

Steps should include:

i. Insert Card
ii. Enter PIN
iii. Validate PIN
iv. Select Withdrawal
v. Enter Amount
vi. Check Balance
vii. Dispense Cash
viii. Print Receipt
ix. Exit

18. Design an algorithm and flowchart to:

Read marks in five subjects, Calculate total, Calculate average , Determine grade , Display Pass/Fail

Rules: Pass if every subject has marks ≥ 40,

Grade: A (≥90) , B (75–89) , C (60–74) , D (40–59) , F (<40)